WASP-10b

R-0879

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_170923` · created 2026-07-29T00:27:49+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458019.6542 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.021
Transit depth (Rp/Rs)^2	2.09 +/- 0.6 [%]
Orbital Inclination (inc)	87.74 +/- 1.05
Airmass coefficient 1 (a1)	0.9771 +/- 0.0091
Airmass coefficient 2 (a2)	0.02 +/- 0.027
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	#1 - [368, 69]
Optimal Aperture	4.11
Optimal Annulus	11.89
Transit Duration (day)	0.0694 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.021 vs 0.1592 (deviation 0.68σ)
Duration fit vs published	0.0694 d vs 0.0946 d (deviation 4.58σ)
Mid-time O–C	8.9 min
Depth SNR	6.9
Residual scatter	0.93 %

Light curve

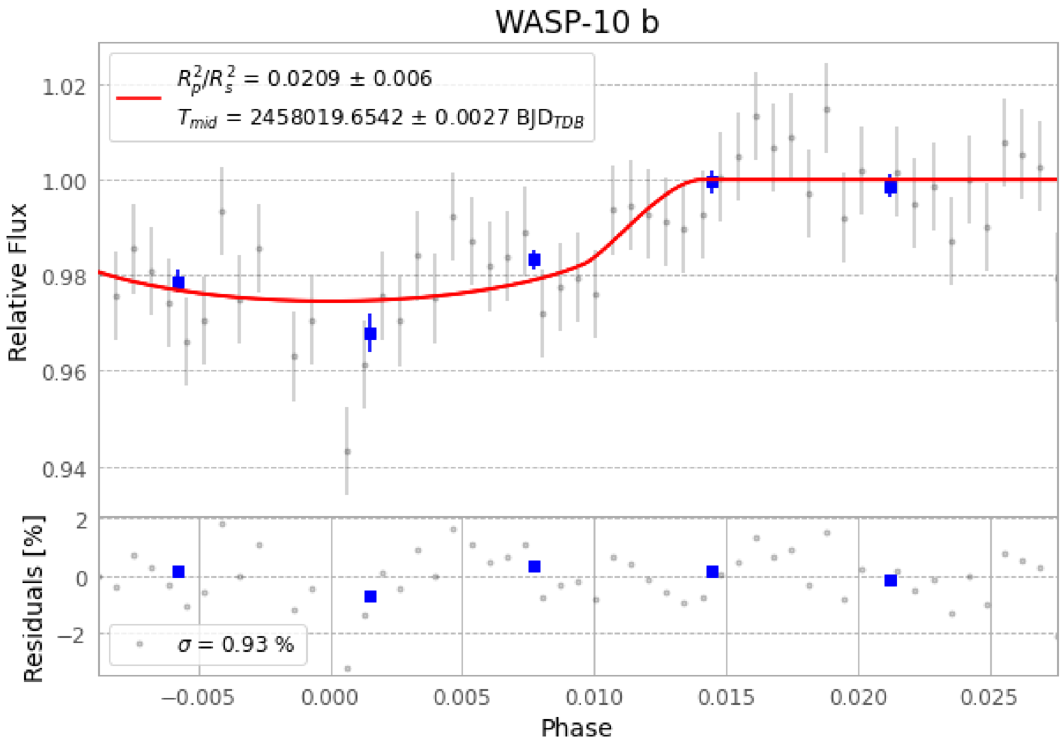

Figure 1 — FinalLightCurve_WASP-10 b_2017-09-22.png (SHA-256 d84dca26e4ad2624…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [492, 98], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 45ca016ec160f30a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

55 FITS frames (36 MB), `WASP-10170923025412.FITS` → `WASP-10170923053612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-170923.log (59030e461529…), FinalLightCurve_WASP-10 b_2017-09-22.png (d84dca26e4ad…), FinalLightCurve_WASP-10 b_2017-09-22.csv (0267e761da00…)

Compute resources

Wall time 345.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)